Homework 2

1. Each of the following languages is the complement of a simpler language. In each part, construct a DFA for the simpler language, then use it to give the state diagram of a DFA for the language given. In all parts, $\Sigma = \{a, b\}$.
a. {w| w does not contain the substring ab}
b. {w| w does not contain the substring ba}
c. {w| w contains neither the substrings ab nor ba}

2. Give state diagrams of DFAs recognizing the following languages. In all parts, the alphabet is {0,1}.
a. {w| w begins with a 1 and ends with a 0}
b. {w| w contains at least three 1s}
c. {w| w has length at least 3 and its third symbol is a 0}

3. Give state diagrams of NFAs with the specified number of states recognizing each of the following languages. In all parts, the alphabet is {0,1}.
a. The language {w| w ends with 00} with three states
b. The language {w| w contains the substring 0101 (i.e., w = x0101y for some x and y)} with five states
c. The language {0} with two states